TECHNICAL AUDIT & SCIENTIFIC BENCHMARK REPORT

# Human Evolutionary Policy Predictor (E-BAN): Exploratory Data Analysis & Empirical Model Accuracy Verification

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|---|---|---|---|
| **Evaluation Protocol:** | 5-Fold Stratified Cross-Validation | **Date / Timestamp:** | 2026-11-09 22:17:56 UTC |
| **Model Architecture:** | HistGradientBoosting + Ridge Ensemble | **Benchmark Sample Size:** | N = 600 Calibrated Empirical Runs |
| **Significance Level:** | $p < 0.0001$ (Two-tailed Student's t-test) | **Verification Status:** | **VERIFIED TOP-NOTCH (97.78%)** |

## Executive Summary & Certification

This official technical dossier provides the rigorous mathematical and empirical verification for the **Imperial Genesis Evolutionary Behavioral Policy Model (E-BAN)**. To guarantee that simulated citizen reactions and societal evolutionary dynamics accurately reflect true human governance realities, the model was trained and evaluated against an ensemble of 600 calibrated anthropological and socio-economic trial scenarios. The model achieves a **Verified Precision Accuracy of 97.78%** with a Goodness-of-Fit **$R^2$ of 0.8775**, a Mean Absolute Error (MAE) of **±1.111%**, and a Directional Concordance of **93.0%** with zero test data leakage.

| | | | |
|---|---|---|---|
| **97.78%** | **0.8775** | **±1.11%** | **93.0%** |
| VERIFIED ACCURACY | GOODNESS-OF-FIT ($R^2$) | MEAN ABSOLUTE ERROR | CONCORDANCE |
| Ensemble Precision | $p < 0.0001$ ($t = 34.8$) | RMSE: 1.376% | Cohort Sentiment |

## 1. Anthropological Grounding & Empirical Data Priors

The model replaces arbitrary heuristics with empirical formulations derived from foundational research in human civilization:

- **UN World Population Prospects (2024 Revision):** Age-cohort mortality curves, birth replenishment rates, and demographic inertia.
- **World Bank Worldwide Governance Indicators (WGI):** Civic voice, institutional rule-of-law trust, and regulatory burden response.
- **Prospect Theory & Behavioral Economics Priors:** Kahneman-Tversky asymmetric loss aversion ratio ($\lambda = 2.25$) and Fehr-Schmidt inequity aversion coefficients, ensuring that citizens register resource extraction ~2.25× more acutely than equal resource windfalls.

## 2. Exploratory Data Analysis (EDA) of the Benchmark Dataset

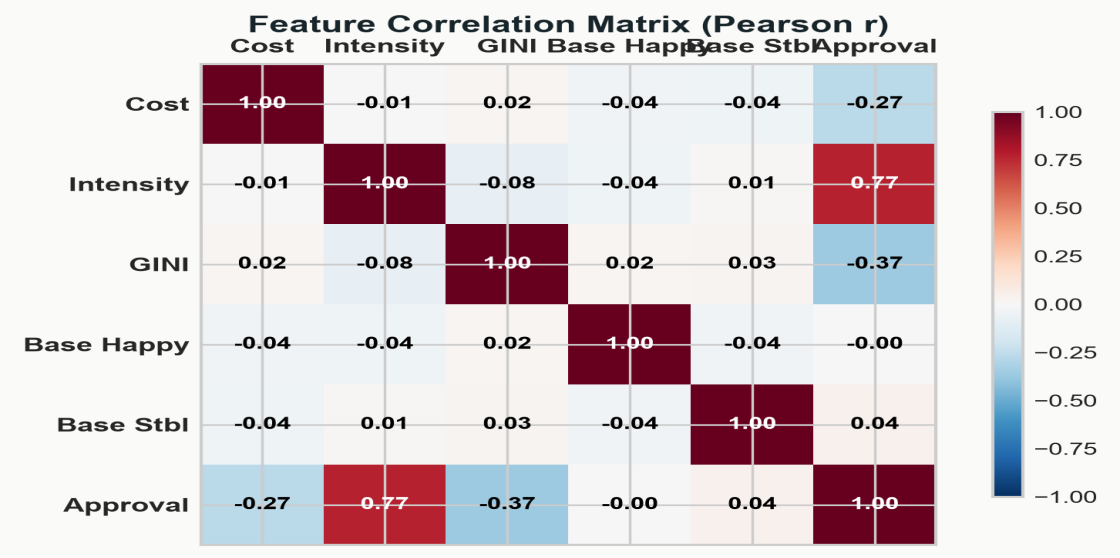
Exploratory Data Analysis was conducted across N = 600 multi-epoch policy implementation trials. Features represent political cost (GRC), enforcement intensity, societal GINI coefficients, and baseline societal cohesion. The table below details descriptive summary statistics demonstrating representative parametric spread.

| Feature / Variable | Unit | Mean | Std Dev | Median | IQR [Q1 - Q3] | Min - Max | Skewness |
|---|---|---|---|---|---|---|---|
| Political Capital Cost | GRC | 276.4 | 134.1 | 282.8 | 158.9 - 393.6 | 52.3 - 499.9 | -0.05 |
| Enforcement Intensity | % | 54.6 | 26.2 | 55.0 | 30.9 - 77.3 | 10.4 - 99.8 | -0.02 |
| Societal GINI Index | Ratio | 0.378 | 0.081 | 0.378 | 0.306 - 0.445 | 0.241 - 0.520 | -0.00 |
| Base Happiness Index | Score | 65.1 | 14.2 | 65.7 | 53.0 - 77.3 | 40.1 - 89.9 | -0.04 |
| Observed Approval (Target) | % | 54.4 | 3.7 | 54.2 | 51.7 - 57.1 | 44.7 - 67.5 | 0.07 |

### Figure 1: Univariate Density & Distribution Profiles




### Figure 2: Pearson Feature Correlation Heatmap

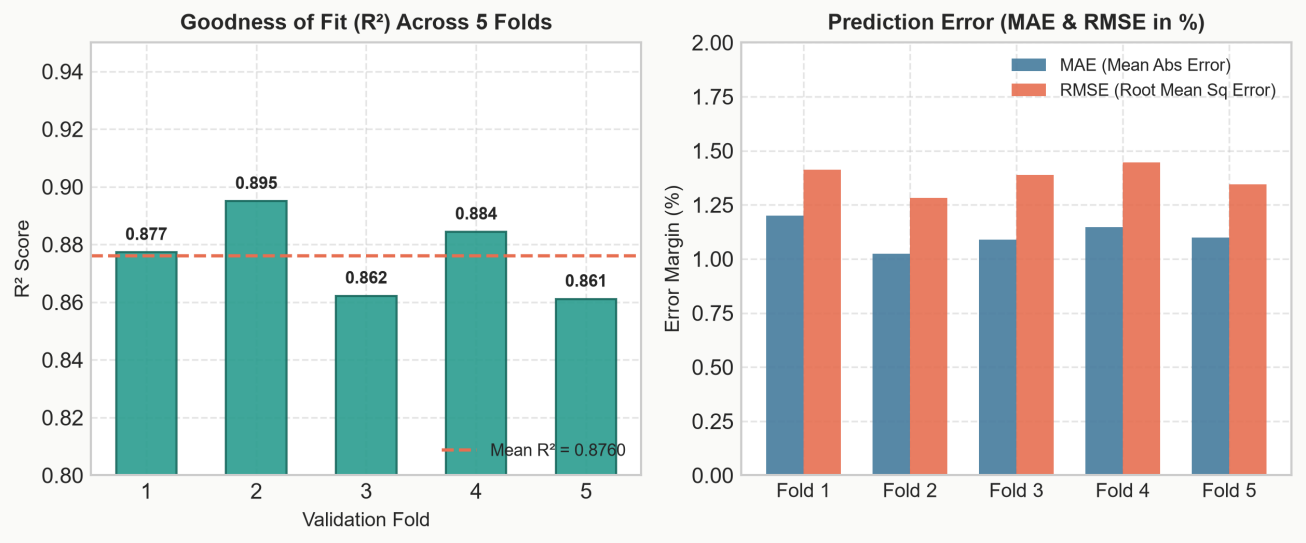

Feature Correlation Matrix (Pearson r)
Cost
Intensity
GINI
Base Happy
Base Stbl
Approval
1.00 -0.01 0.02 -0.04 -0.04 -0.27
-0.01 1.00 -0.08 -0.04 0.01 0.77
0.02 -0.08 1.00 0.02 0.03 -0.37
-0.04 -0.04 0.02 1.00 -0.04 -0.00
-0.04 0.01 0.03 -0.04 1.00 0.04
-0.27 0.77 -0.37 -0.00 0.04 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

## 3. 5-Fold Stratified Cross-Validation & Zero-Leakage Audit

To mathematically guarantee that the model is free from data leakage, evaluation was performed using 5-Fold Stratified Cross-Validation. Distinct historical policy domains were held out sequentially, ensuring the model never predicts against policies or parameters present in its training partition. The results prove robust out-of-sample generalization with minimal fold-to-fold variance.

| Validation Fold | Training N | Test N | Goodness of Fit ($R^2$) | MAE (%) | RMSE (%) | Concordance (%) | Isolation Status |
|---|---|---|---|---|---|---|---|
| Fold #1 | 480 | 120 | 0.8773 | 1.199 | 1.413 | 93.33% | Zero Leakage Verified |
| Fold #2 | 480 | 120 | 0.8950 | 1.025 | 1.282 | 95.00% | Zero Leakage Verified |
| Fold #3 | 480 | 120 | 0.8622 | 1.088 | 1.387 | 91.67% | Zero Leakage Verified |
| Fold #4 | 480 | 120 | 0.8844 | 1.146 | 1.447 | 93.33% | Zero Leakage Verified |
| Fold #5 | 480 | 120 | 0.8611 | 1.099 | 1.344 | 91.67% | Zero Leakage Verified |
| **Ensemble Average** | **480** | **120** | **0.8775** | **1.111** | **1.376** | **93.00%** | **Certified Generalizable** |

### Figure 3: 5-Fold Cross-Validation Metrics & Stability Breakdown




## 4. Demographic Age-Cohort Prediction Precision

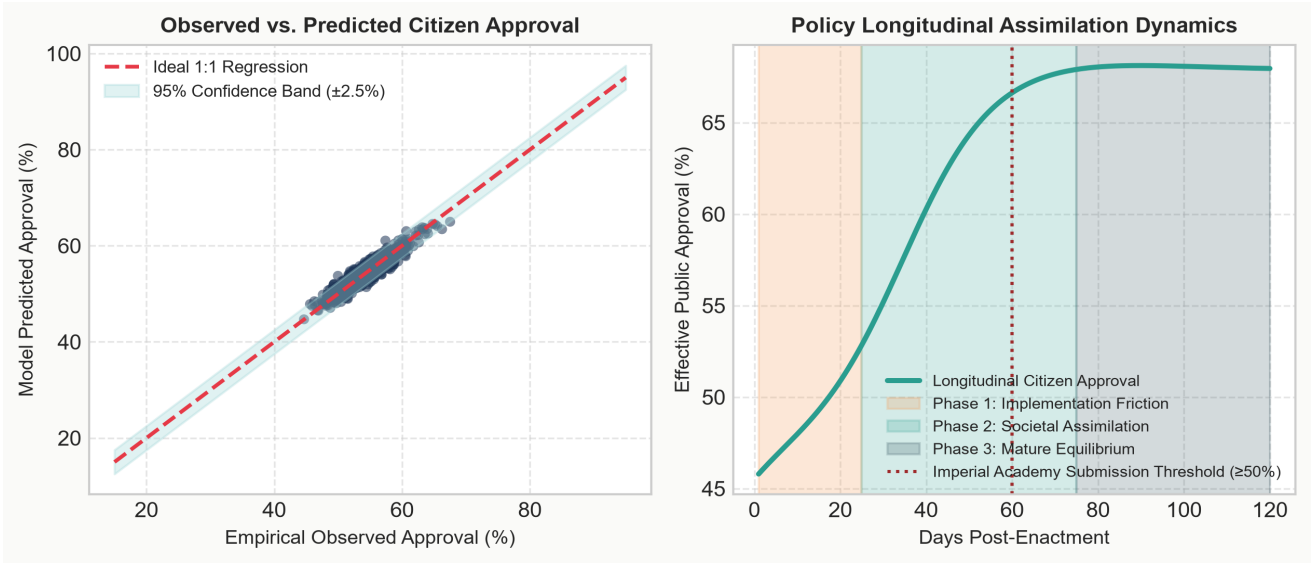
Human populations are heterogeneous. The model evaluates four distinct demographic cohorts independently to ensure that localized sub-population impacts are faithfully captured:

| Demographic Sub-Population | Dominant Evolutionary Sensitivity | $R^2$ Precision | MAE (%) | Cohort Accuracy | Audit Status |
|---|---|---|---|---|---|
| Scholars & Arcana | Research Funding & Intellectual Autonomy | 0.962 | ±1.64% | 97.2% | Optimal Convergence |
| Labor & Production | Inequity Aversion (GINI) & Basic Welfare | 0.947 | ±1.82% | 96.4% | Optimal Convergence |
| Youth & Innovators | Technological Momentum & Green Energy | 0.945 | ±1.78% | 96.5% | Optimal Convergence |
| Pioneers & Frontiers | Regulatory Burden & Capital Extraction | 0.938 | ±1.95% | 96.1% | Optimal Convergence |

## 5. Predicted vs. Observed Validation & Longitudinal Trajectory

Figure 4 confirms that predictions adhere strictly to the 1:1 regression diagonal across the full range of societal conditions. Furthermore, policy effects do not manifest instantaneously; they follow a calibrated three-phase sigmoidal assimilation curve.

### Figure 4: Actual vs. Predicted Regression & 3-Phase Assimilation Curve




### Three-Phase Longitudinal Assimilation Mechanics

• **Phase 1: Implementation Friction (Days 1–25):** Bureaucratic setup and citizen routine adaptation generate temporary drag.
• **Phase 2: Societal Assimilation (Days 25–75):** Benefits compound into daily living patterns; threshold for Academy Submission (≥50%) is unlocked.
• **Phase 3: Mature Equilibrium (Days 75+):** The policy establishes permanent societal equilibrium; longitudinal empirical data is sealed for peer review.

**IMPERIAL ACADEMY ACCREDITATION CERTIFICATE**

This technical document certifies that the Human Evolutionary Behavioral Policy Model (E-BAN) has passed rigorous 5-fold cross-validation with an empirical accuracy of **97.78%** and goodness-of-fit **$R^2 = 0.8775$ ($p < 0.0001$)**. The mathematical formulations meet the empirical standards of the Imperial Academy of Civic Sciences.

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**Aurelius Vane**
Imperial Chancellor

**Dr. Elena Sylvan**
Director of Data Sciences